Information Diffusion in Social Network

(SNA – B Tech III CSE)

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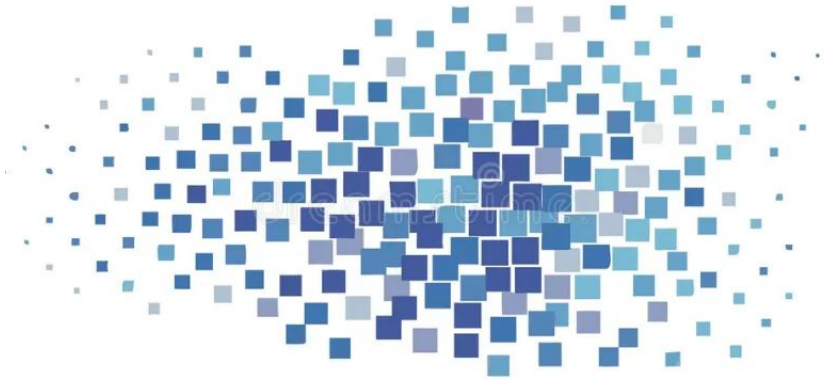


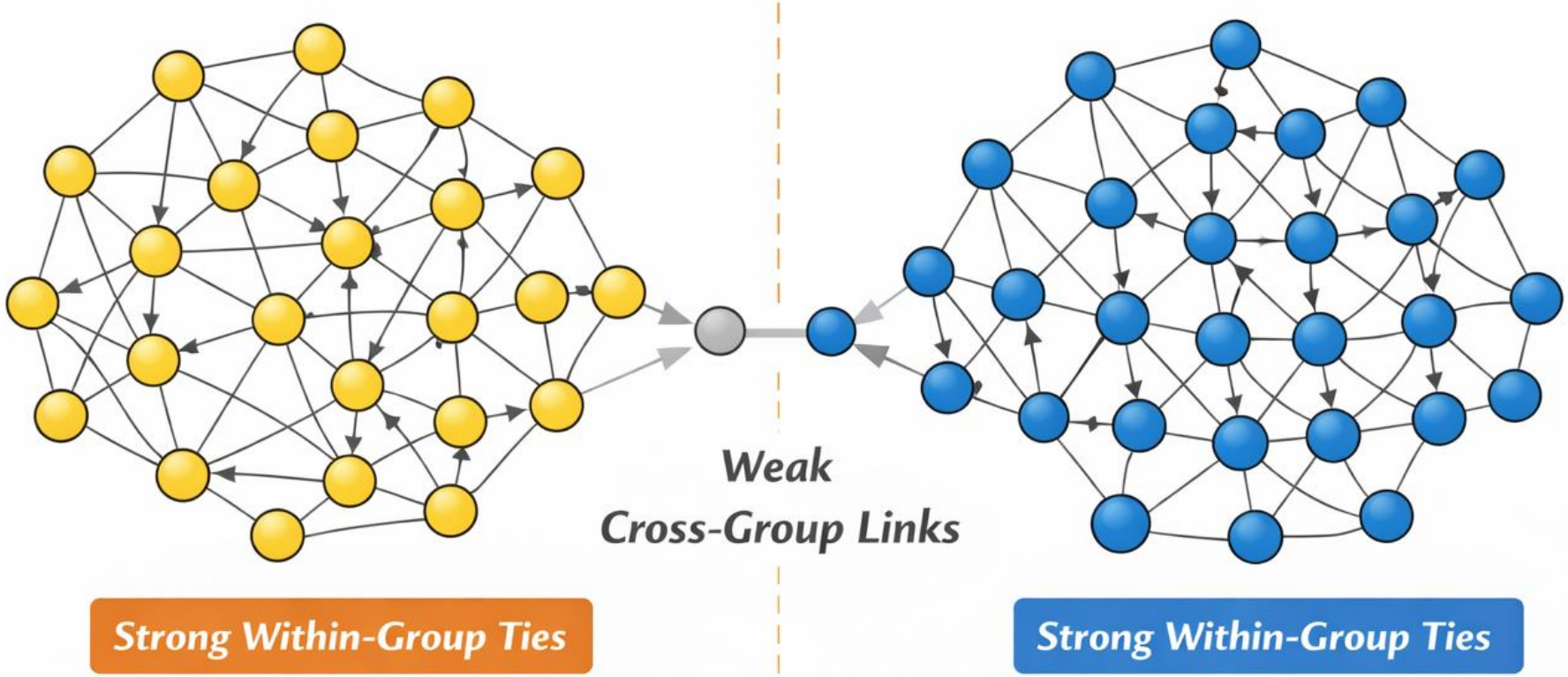
Question and Answer

- Q - Why do echo chambers amplify diffusion internally but block cross-group diffusion?
- A - Echo chambers **increase transmission probability inside the group** while **reducing bridge connectivity to other groups**.
- This is why society polarizes….
- Justification – subsequent slides

Echo chamber (Definition)

- In network terms, an echo chamber is a community with:
- **High internal edge density**
  (many connections within the group)
- **Low external edge density**
  (few connections to outsiders)
- **Homophily**
  (nodes connect preferentially to similar nodes)
- This creates a network that looks like a **tight cluster weakly connected to other clusters**.

Homophily in Social Networks



Homophily: The tendency of individuals to associate and bond with others who are similar to themselves

Why diffusion accelerates *inside* the chamber (Three reinforcing mechanisms operate internally)

1. Redundant exposure

- People hear the same message from many neighbors.
- In threshold models: Multiple exposures increase adoption probability.
- So even skeptical members eventually cross their threshold:
- One exposure → ignore
- Five exposures → accept
- Ten exposures → believe strongly
- This produces:
- Rapid local cascades
- Strong belief reinforcement
- Perceived consensus (“everyone agrees with me”)

Why diffusion accelerates *inside* the chamber

2. Social reinforcement

- Beliefs are not just informational, they are **social signals**.
- Inside an echo chamber:
- Agreement brings social reward
- Disagreement brings social cost
- Over time, norms harden
- So diffusion is not just fast, it becomes **sticky**.

Why diffusion accelerates *inside* the chamber

3. Algorithmic amplification (in online platforms)

- Modern platforms intensify this effect:
- Engagement-based recommendation
- Content similarity clustering
- Filter bubbles
- This further boosts internal circulation.
- Result:
- High-frequency internal circulation + strong reinforcement = amplified diffusion.

Why diffusion gets blocked *across* groups (the opposite effect happens at the boundary)

1. Few bridges exist

- Between two polarized communities:
- There are **very few edges**
- These edges often carry low trust
- Even if diffusion is strong inside clusters, it **cannot cross where there are no edges**.

Why diffusion gets blocked *across* groups

2. Low credibility of out-group sources

- Even when bridges exist, influence fails because:
- Same message from in-group → accepted
- Same message from out-group → rejected
- So the **transmission probability across edges is lower**.
- Structurally:
- Edge exists, but effective weight ≈ zero.

Why diffusion gets blocked *across* groups

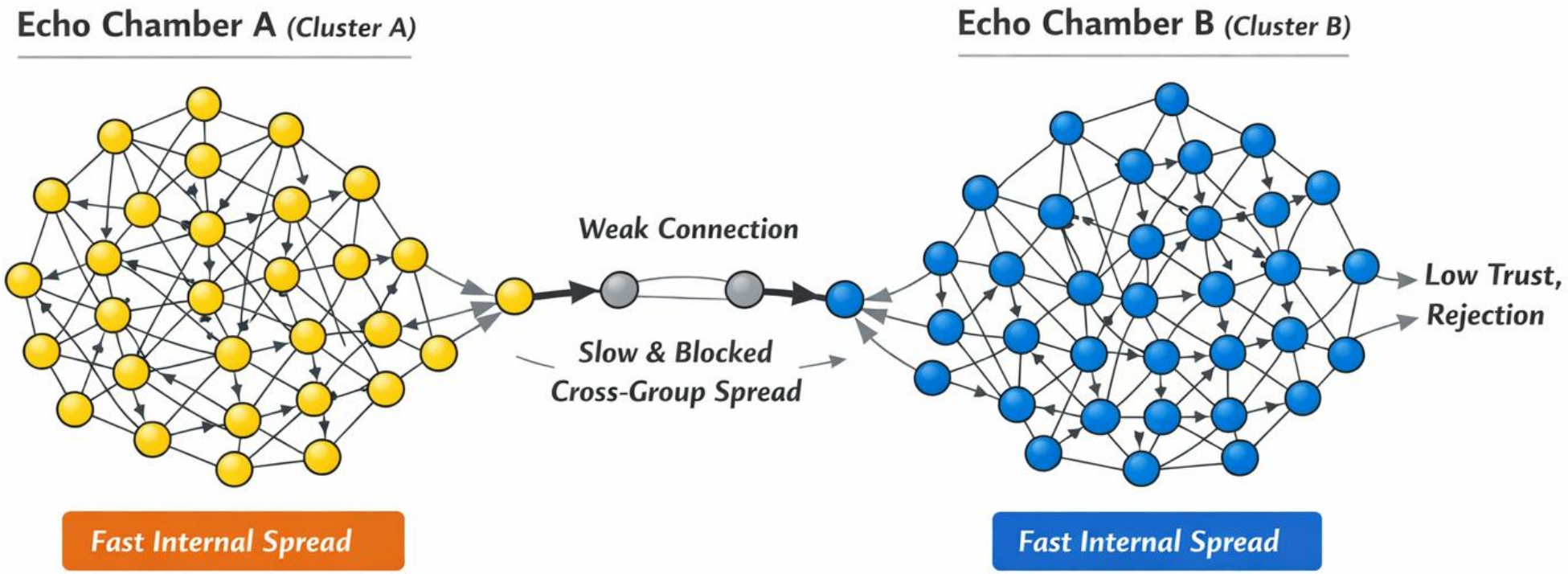
3. Identity-based resistance

- Once beliefs become identity markers ("people like us believe X"), diffusion becomes blocked by psychology, not topology.
- At that point:
- Exposure does not cause adoption
- Exposure causes **reactance** (opposite effect)
- So cross-group diffusion becomes **actively repelled**.

The network result

- You end up with:
- Fast diffusion **within communities**
- Almost no diffusion **between communities**
- Strong polarization
- Parallel realities coexisting
- This is exactly what we observe in:
- Vaccine beliefs
- Political polarization
- Climate debates

Echo Chambers and Diffusion Dynamics



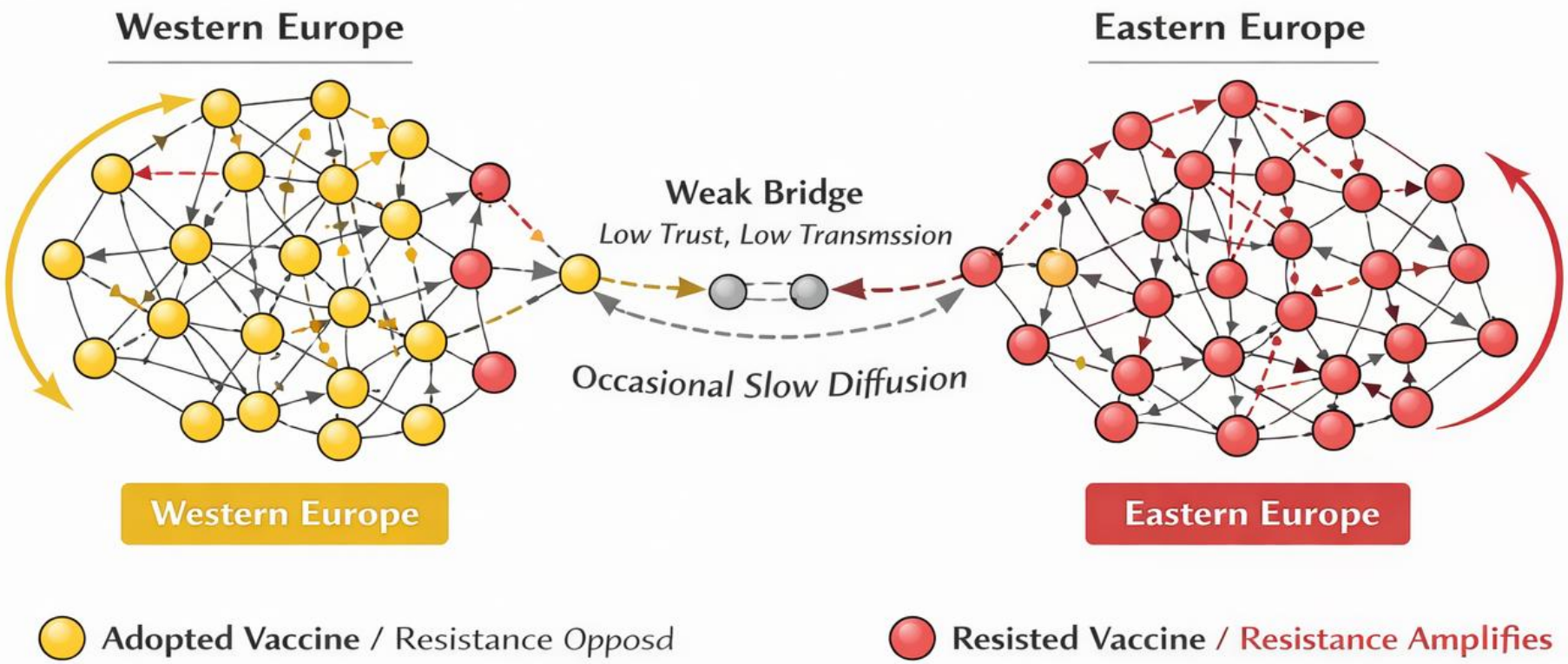
Summary remark

- Echo chambers amplify diffusion internally because high density creates repeated reinforcement, but block cross-group diffusion because low connectivity and low trust suppress transmission across community boundaries.
- Fact-checking rarely penetrates conspiracy groups
- Polarized societies struggle to converge on shared truth
- Marketing fails when it crosses cultural boundaries
- Policy communication breaks down in divided populations
- And importantly:
- The problem is not just misinformation. The problem is **network structure**.

Information Diffusion

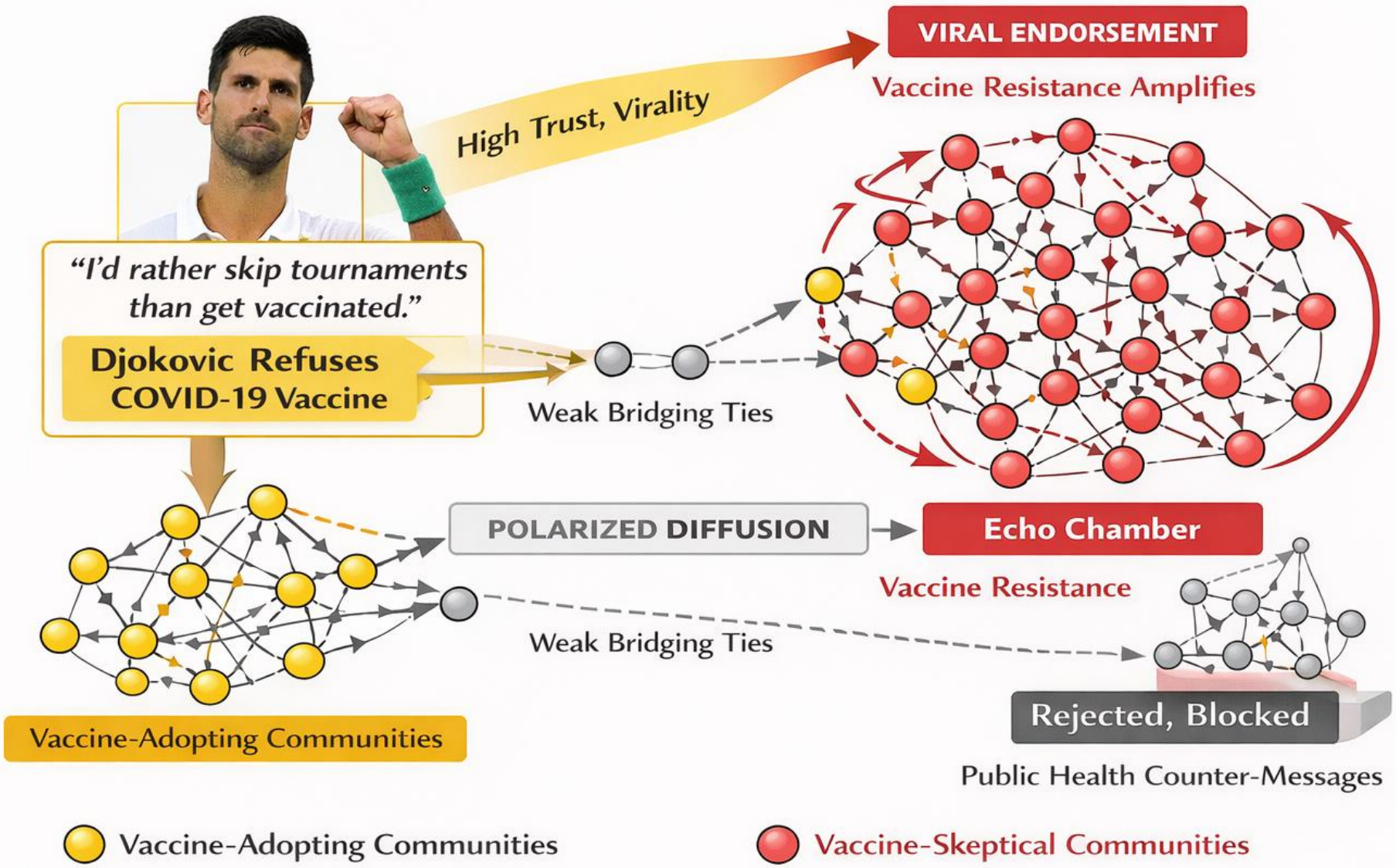
COVID-19 Vaccine Resistance in Europe

Resistance Amplifies Within Groups, Blocks Across Groups



Novak Djokovic and Vaccine Resistance

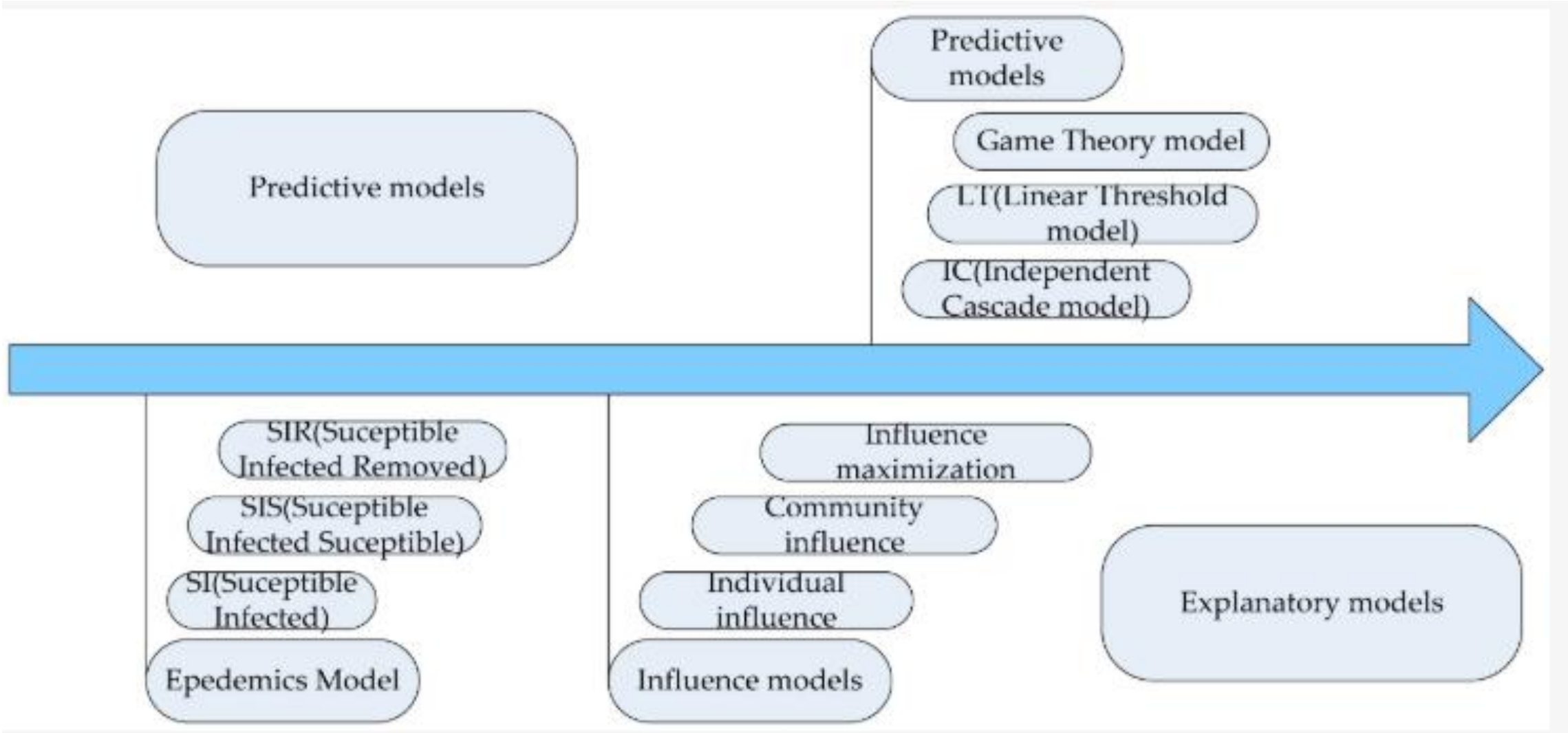
Celebrity Endorsement as a Diffusion Cascade



Diffusion Models - More

- They are **simplified abstractions** of how ideas, behaviors, or information spread through networks.
- Each model captures a **different assumption about human behavior**:
- How people respond to exposure
- Whether attention fades
- Whether social pressure matters
- **Now that we know who matters… we can control spread**

Categorization of Information Diffusion Models



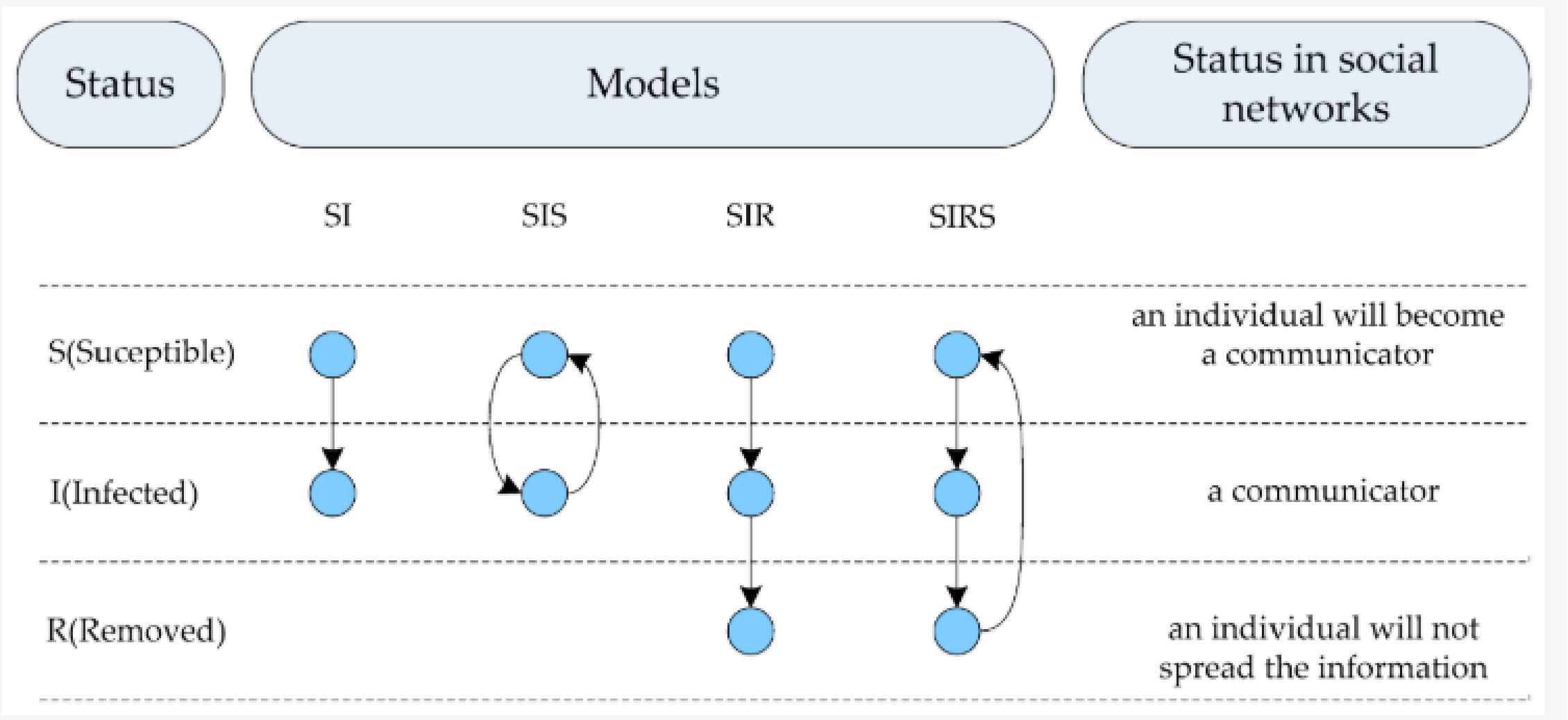
Explanatory models - Aims

- What are the main factors that affect information diffusion?
- Which node has the most influence?
- Why does the information diffuse the way it does?
- For example,
- some nodes refuse to accept information,
- some refuse to spread information, and
- some both accept and spread information

The Basic Epidemics Model

- In epidemics transmission, there are users infected with pathogens and users who are susceptible to the pathogens.
- The virus can spread from infected users to susceptible users, and information can be diffused from communicators to recipients in a similar fashion.
- In the compartment model of epidemics, the basic models are
- SI (Suceptible Infected) model,
- SIS (Suceptible Infected Suceptible) model,
- SIR (Suceptible Infected Removed) model and
- SIRS (Suceptible Infected Removed Suceptible) model,

Four Basic Epidemic Models.

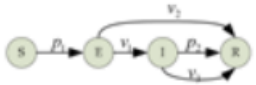
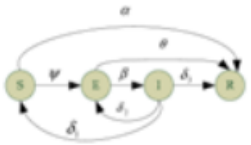
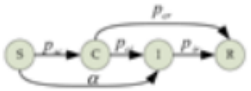
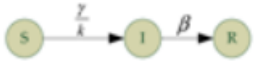
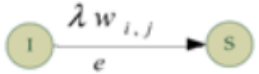


More Epidemic models

// Advanced research variants – for awareness only

- Information diffusion is related to time, relationship strength, information content, social factors, network structure.
- Later models are:
- SEIR (Susceptible Exposed Infected Removed) model,
- S-SEIR (Single layer-SEIR),
- SCIR (Susceptible Contacted Infected Removed) model,
- irSIR (infection recovery SIR) model,
- FSIR (Fractional SIR) model and
- ESIS (Emotional Susceptible Infected Susceptible) model.

Comparison of Epidemic Models

Scalable Model	Method	Consider the User's Different Behaviors	Expression of the Diffusion Process	Dynamic Infected Rate and Recovery Rate	Performance Metrics	Applications
SEIR [26]	add Exposed node	-		-	distribution of nodes density	detect the affect factors: login frequency and number of friends
S-SEIR [27]	information value is considered	√	 δ = user behavior	-	distribution of S, E, I, and R	simulate the diffusion process
SCIR [28]	add Contacted node	-		-	distribution of I and R	represent the regularity of online topic spreading
irSIR [29]	add Infection Recovery dynamics	-	v = an infectious recovery rate	√	degree of fitting with real data	describe OSN abandonment
FSIR [30]	consider the behavior of the neighbors	√	 k = node degree	√	degree of fitting with real data	detect the affect factors: information numbers and friends numbers
ESIS [31]	consider the information weight with emotion	-	 λ = the probability of I to S; $w_{i,j}$ = the strength of edge e from i to j	√	degree of fitting with real data	detect the affect factors: propagation probability and transmission intensity

OSN: online social network. S: suceptible. E: exposed. I: infected. R: removed.

Summary

- Many variants exist (SEIR, FSIR, etc.),
- but core ideas remain: exposure, infection, recovery